

Srikrishnaa Vadivel | Machine Learning Engineer | Scientific ML | Equivariant Models

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Profile

PhD researcher applying machine learning to first-principles materials and molecular systems. Experience spans PyTorch, equivariant graph networks, diffusion models, distributed training concepts, HPC data pipelines, and physics-based loss design.

ML Experience

Yale University

PhD Researcher

2021–present

- Developed first-principles datasets and workflows for self-trapped exciton calculations on leadership-class HPC resources.
- Prototyped EGNN-style force models with energy gradients $F_{ai} = -\partial E / \partial R_{ai}$ and molecular graph construction.
- Built diffusion-model notes and code scaffolds covering DDPM training, conditional inference, transformers, and index-notation derivations.

Probe Technologies / Weatherford

Software and Embedded Engineer

2019–2021

- Built GPU-accelerated processing and visualization for large acoustic waveform data streams using CUDA and OpenGL.
- Engineered production data tooling around ASP services and Microsoft database systems.

Selected Projects

EGNN QM9 Force Model: PyTorch message-passing model, force loss, synthetic graph fallback, and training-curve generation.

Diffusion Notes: DDPM equations with α_t , β_t , conditional training, and inference in index notation.

LiF Band Scaffold: Reciprocal-space Hamiltonian assembly and eigenvalue plotting for physics-informed ML context.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BA, Electrical and Computer Engineering

2013–2017

Skills

ML: PyTorch, scikit-learn, Hugging Face Transformers, Accelerate, datasets

Scientific: NumPy, SciPy, pandas, PETSc, SLEPc

Systems: Python, C++, Linux, Docker, Kubernetes, MPI, OpenMP

Compute: CUDA